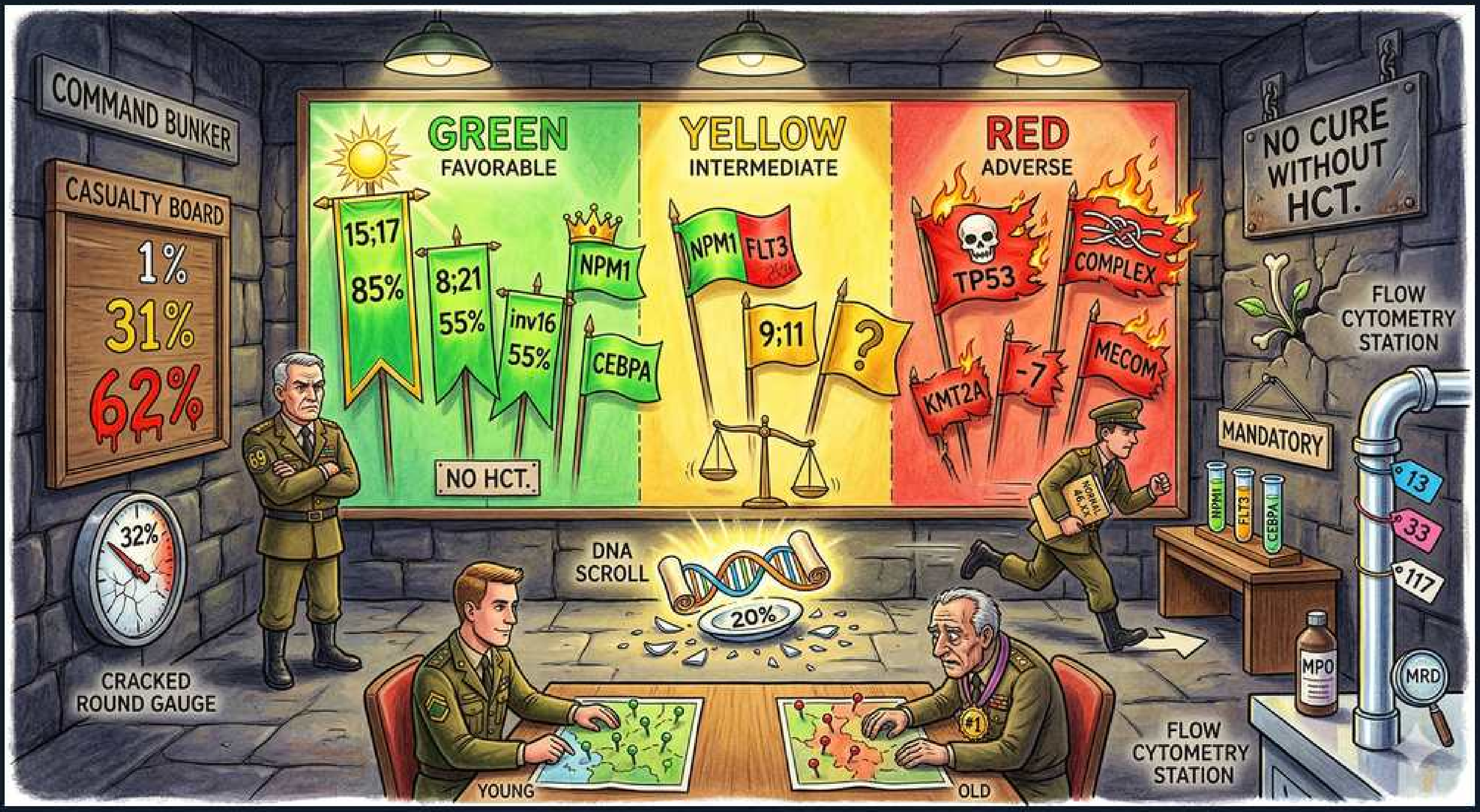
AML — ELN Risk Stratification (Green / Yellow / Red)



1. GREEN — Favorable

WHAT YOU SEE

Green panel headed 'GREEN — FAVORABLE'.

WHAT IT ENCODES

Favorable ELN group: best survival, treated with chemotherapy consolidation and NO allo-HCT in CR1. Contains t(15;17)/APL, CBF AMLs, NPM1-mut without high FLT3-ITD, and bZIP CEBPA.

BOARD TRAP

Favorable does not equal cured — relapse still occurs, so MRD monitoring continues. But upfront transplant is avoided.

2. 15:17 / 85% APL

WHAT YOU SEE

'15:17 85%' on the green flag.

WHAT IT ENCODES

t(15;17) PML-RARA = acute promyelocytic leukemia, the most favorable AML, ~85%+ cure with ATRA + arsenic trioxide. A hematologic emergency due to DIC.

BOARD TRAP

APL is favorable in prognosis but a true emergency at presentation — start ATRA on suspicion before genetics confirm, because of fatal coagulopathy.

3. 8;21 & inv16 55%

WHAT YOU SEE

'8;21 55%' and 'inv16 55%' on green flags.

WHAT IT ENCODES

Core-binding factor AMLs (t(8;21) RUNX1-RUNX1T1; inv(16) CBFB-MYH11). Favorable, chemo-sensitive, benefit from high-dose cytarabine consolidation.

BOARD TRAP

Both are CBF and favorable — don't split them into different risk tiers. KIT mutation can downgrade outcome within CBF.

4. NPM1 / CEBPA tiles

WHAT YOU SEE

Green tiles 'NPM1' and 'CEBPA'.

WHAT IT ENCODES

NPM1-mutated (without high-ratio FLT3-ITD) and in-frame bZIP CEBPA are favorable normal-karyotype AMLs. NPM1 is also a sensitive PCR MRD target.

BOARD TRAP

NPM1 is favorable only when FLT3-ITD is absent/low. On the yellow panel, NPM1 reappears beside FLT3 — that combination is intermediate.

5. YELLOW — Intermediate

WHAT YOU SEE

Yellow panel 'YELLOW — INTERMEDIATE' with balance scale.

WHAT IT ENCODES

Intermediate ELN group: allo-HCT in CR1 is decided case-by-case using MRD, comorbidity, and donor availability. Includes NPM1+FLT3-ITD combinations and abnormalities not classified favorable or adverse.

BOARD TRAP

Intermediate is the 'it depends' tier — MRD status after induction often tips the decision toward or away from transplant.

6. NPM1 + FLT3 9;11

WHAT YOU SEE

'NPM1', 'FLT3', '9;11', and a '?' on yellow.

WHAT IT ENCODES

NPM1-mut with FLT3-ITD, plus t(9;11) KMT2A (MLLT3), sit in the intermediate band. Add a FLT3 inhibitor (midostaurin) when FLT3 is mutated.

BOARD TRAP

t(9;11) is the one KMT2A rearrangement classed intermediate; OTHER KMT2A rearrangements (e.g. t(6;11), t(11;19)) are adverse — see the red panel.

7. Balance scale

WHAT YOU SEE

Scales of justice on the yellow panel.

WHAT IT ENCODES

Symbolizes the weighing of transplant risk vs benefit in intermediate disease. The scale tips on residual disease and fitness.

BOARD TRAP

Don't auto-transplant or auto-spare intermediate patients; the answer is individualized risk assessment.

8. RED — Adverse

WHAT YOU SEE

Red flaming panel 'RED — ADVERSE'.

WHAT IT ENCODES

Adverse ELN group: poor survival, allo-HCT in CR1 is the standard if the patient is fit. Includes TP53, complex/monosomal karyotype, adverse KMT2A, and certain other lesions.

BOARD TRAP

Adverse risk = pursue transplant in first remission. Treating adverse-risk AML with chemo alone and skipping transplant is the wrong move in a fit patient.

9. TP53

WHAT YOU SEE

Red 'TP53' tag with flames.

WHAT IT ENCODES

TP53 mutation (especially multi-hit or with complex karyotype) is the worst AML prognosis; resistant to chemo and even to transplant. Often associated with therapy-related and MDS-related AML.

BOARD TRAP

TP53 AML responds poorly to intensive 7+3; venetoclax/HMA or clinical trials are often preferred, and outcomes remain dismal even post-transplant.

10. Complex karyotype

WHAT YOU SEE

Red 'COMPLEX' tag.

WHAT IT ENCODES

Complex (≥3 abnormalities) and monosomal karyotypes are adverse. Frequently overlap with TP53 and prior MDS/therapy exposure.

BOARD TRAP

Complex karyotype is adverse regardless of an otherwise favorable lesion — a coexisting complex karyotype overrides a single favorable marker.

11. KMT2A / adverse

WHAT YOU SEE

Red 'KMT2A' tag.

WHAT IT ENCODES

KMT2A (MLL, 11q23) rearrangements other than t(9;11) are adverse, as are abnormalities like t(6;9) DEK-NUP214 and inv(3)/t(3;3) MECOM.

BOARD TRAP

Only t(9;11) KMT2A is intermediate; the generic '11q23 / KMT2A rearranged' on the red panel is adverse — direction matters.

12. MECOM / inv(3)

WHAT YOU SEE

Red 'MECOM' tag.

WHAT IT ENCODES

inv(3)/t(3;3) activating MECOM (EVI1) is adverse, often with normal/high platelets and multilineage dysplasia. Poor response to standard therapy.

BOARD TRAP

inv(3) AML can present with normal or elevated platelet counts — a clue that distinguishes it from most AMLs that present with low counts.

13. FLT3 (red)

WHAT YOU SEE

'FLT3' near the red panel.

WHAT IT ENCODES

High-allelic-ratio FLT3-ITD pushes toward adverse. Pairing with NPM1 modifies risk; FLT3 inhibitors (midostaurin frontline, gilteritinib at relapse) are added.

BOARD TRAP

Gilteritinib is the relapsed/refractory FLT3 drug; midostaurin is the frontline add-on. Swapping them is a classic distractor.

14. Casualty board 1/31/62%

WHAT YOU SEE

'CASUALTY BOARD 1% / 31% / 62%' rising mortality.

WHAT IT ENCODES

Stylized increasing mortality across favorable→intermediate→adverse tiers, reinforcing that risk group drives both prognosis and the transplant decision.

BOARD TRAP

These are tiered outcome figures, not a single survival number — the gradient is the point: risk tier predicts survival.

15. Flow cytometry / MRD

WHAT YOU SEE

'FLOW CYTOMETRY STATION' and 'MRD' with MPO.

WHAT IT ENCODES

Flow cytometry confirms myeloid lineage (MPO+) and tracks MRD. MRD status refines the transplant decision, especially in the intermediate group.

BOARD TRAP

Flow MRD complements but is less sensitive than PCR for NPM1/CBF — a PCR-positive, flow-negative patient is still MRD-positive.